Sprint 2 – Eco Sharing

Module: Software Engineering (CMP-N204-0)

Theme: Sharing, exchange, and building community

Group Name: Eco Share

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1. Project Summary (Refined after Sprint 1)

Eco Share is a community-based web application designed to support sustainable and cost-effective living by enabling people to lend and borrow everyday items instead of purchasing new ones. The platform primarily focuses on students, who often need access to items such as textbooks, kitchen equipment, or household tools for short-term use and may have limited budgets.

The idea is informed by common student experiences, including those of international students like me and my members of the group, who may face additional financial and practical challenges when adapting to a new environment. However, Eco Share is designed to be inclusive and allows any local community member to participate in item sharing.

The system allows registered users to list items, browse and search for available items nearby, and request to borrow items for a specified period.

Item owners can approve or reject requests, ensuring accountability, trust, and user safety.

Eco Share is developed using an Agile methodology, with the Minimum Viable Product (MVP) focusing on core functionality such as user authentication, item listing creation,

item discovery, and the borrow-request approval process. These features provide a complete sharing cycle and a foundation for future system enhancements.

Eco Share is intended for the sharing of everyday items such as textbooks, kitchen equipment, household tools, and small appliances for short-term personal use. The platform does not support the sharing of illegal, prohibited, or high-risk items.

This Sprint 2 focuses on refining system requirements and design artefacts to ensure the application is ready for implementation in **Sprint 3**.

2. User Stories

This section presents the prioritised user stories for the Eco Share application. The user stories describe system functionality from the perspective of different users and are aligned with the Minimum Viable Product (MVP) planned for Sprint 3.

Guest User Stories

US-01: Register Account

Priority: High (MVP)

As a guest, I want to register an account so that I can borrow and lend items.

Acceptance Criteria:

- Guests can enter their name, email, and password
- Email must be unique
- Account is created successfully

US-02: Login

Priority: High (MVP)

As a guest, I want to log in so that I can access my account and platform features.

Acceptance Criteria:

- Guests can log in using valid credentials
- Invalid credentials show an error message
- Successful login redirects to home page

US-03: Browse Items

Priority: High (MVP)

As a guest, I want to browse available items so that I can see what items are available before registering.

Acceptance Criteria:

- Guests can view a list of available items
- Basic item details are visible
- Borrowing requires a login

US-04: View Item Details

Priority: High (MVP)

As a guest or registered user, I want to view detailed information about an item so that I can decide whether to request it.

Acceptance Criteria:

- Item details include description and availability
- Item owner information is displayed

Registered User Stories

US-05: Create Item Listing

Priority: High (MVP)

As a registered user, I want to create an item listing so that others can borrow my item.

Acceptance Criteria:

- User enters item title, description, category, and availability
- Listing is saved successfully
- Listing appears in browse items

US-06: Request to Borrow Item

Priority: High (MVP)

As a registered user, I want to request to borrow an item so that I can use it for a specific time period.

Acceptance Criteria:

- User selects borrow start and end dates
- User cannot request their own item
- Request status is set to “Pending.”

US-07: View Borrow Request Status

Priority: High (MVP)

As a registered user, I want to view the status of my borrow requests so that I know whether they are approved or pending.

Acceptance Criteria:

- User can view request status
- Status updates when approved or rejected

US-08: Approve or Reject Borrow Request

Priority: High (MVP)

As an item owner, I want to approve or reject borrow requests so that I can control who borrows my item.

Acceptance Criteria:

- Owner can view incoming requests
- Owner can approve or reject a request
- Request status updates correctly

US-09: View Pickup Details

Priority: Medium (MVP)

As a borrower, I want to view confirmed pickup details so that I know when and where to collect the item.

Acceptance Criteria:

- Pickup details are visible only after approval
- Details are agreed through in-app messaging

US-10: In-App Messaging

Priority: Medium (MVP)

As a registered user, I want to message another user so that pickup details can be discussed safely.

Acceptance Criteria:

- Messaging is enabled after request approval
- Messages remain within the platform

US-11: Report User or Listing

Priority: Medium

As a registered user, I want to report a user or listing so that unsafe behaviour can be reviewed.

Acceptance Criteria:

- User can submit a report with a reason
- Report is stored for admin review

Admin User Stories

US-12: Admin Login

Priority: Medium

As an admin, I want to log in so that I can manage the platform securely.

Acceptance Criteria:

- Admin login requires valid credentials
- Admin-only access is enforced

US-13: Review Reports

Priority: Medium

As an admin, I want to review reported users or listings so that appropriate moderation action can be taken.

Acceptance Criteria:

- Admin can view submitted reports
- Admin can take action on reports

3. Use Case Diagram

The use case diagram illustrates the high-level interactions between different types of users and the Eco Share system. It identifies the main system functionalities available to Guests, registered Users, and Administrators, and provides an overview of how users interact with the platform.

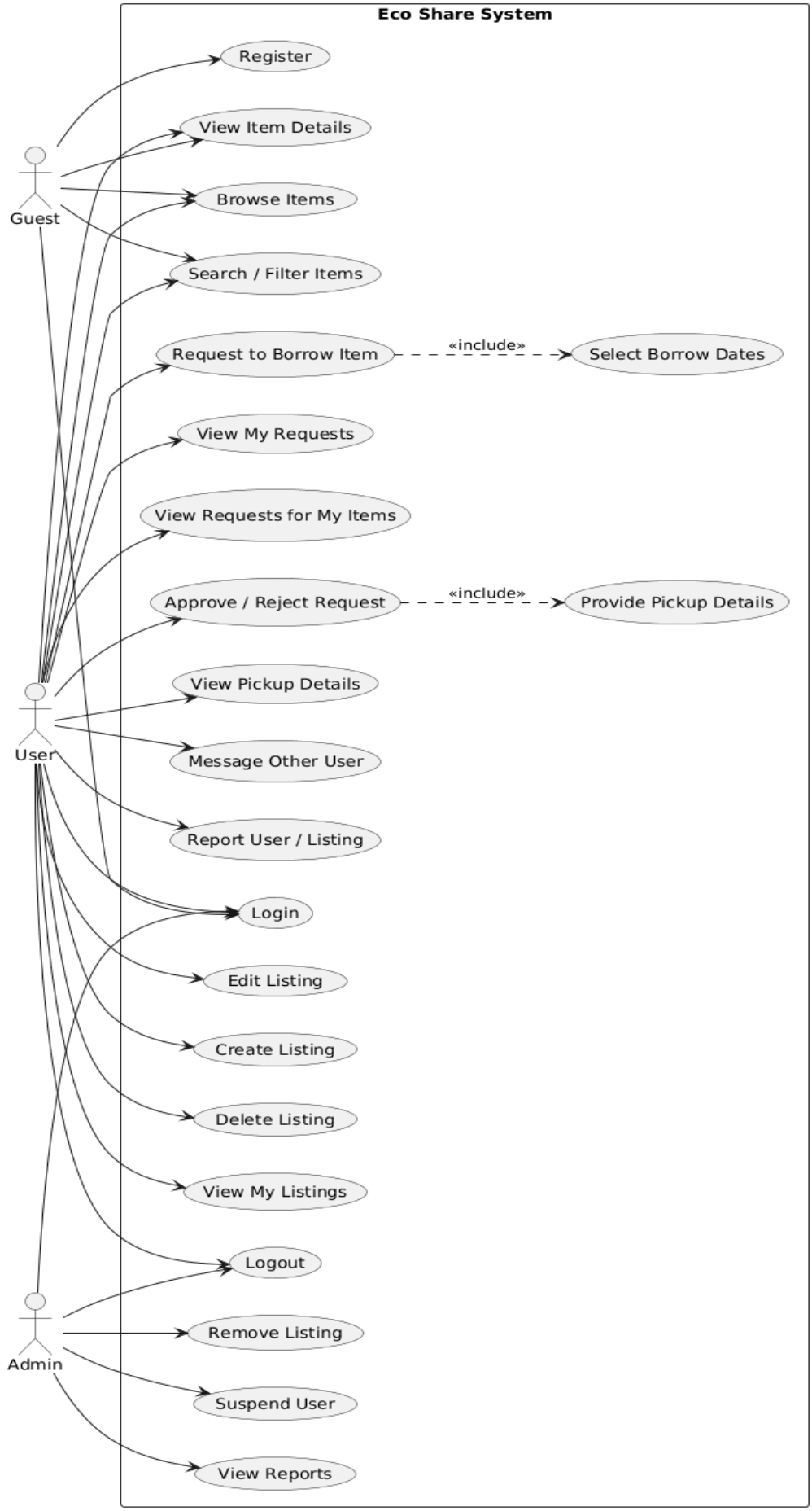


Figure 1: Eco Share Use Case Diagram

This diagram mainly focuses on high-level user goals, while detailed system behaviour is described using flowcharts in the following sections.

5. Wireframes

This section presents low-fidelity wireframes for the Eco Share application. The wireframes represent the key screens required for the Sprint 3 Minimum Viable Product (MVP) and are designed to illustrate layout, navigation, and core user interactions rather than final visual design.

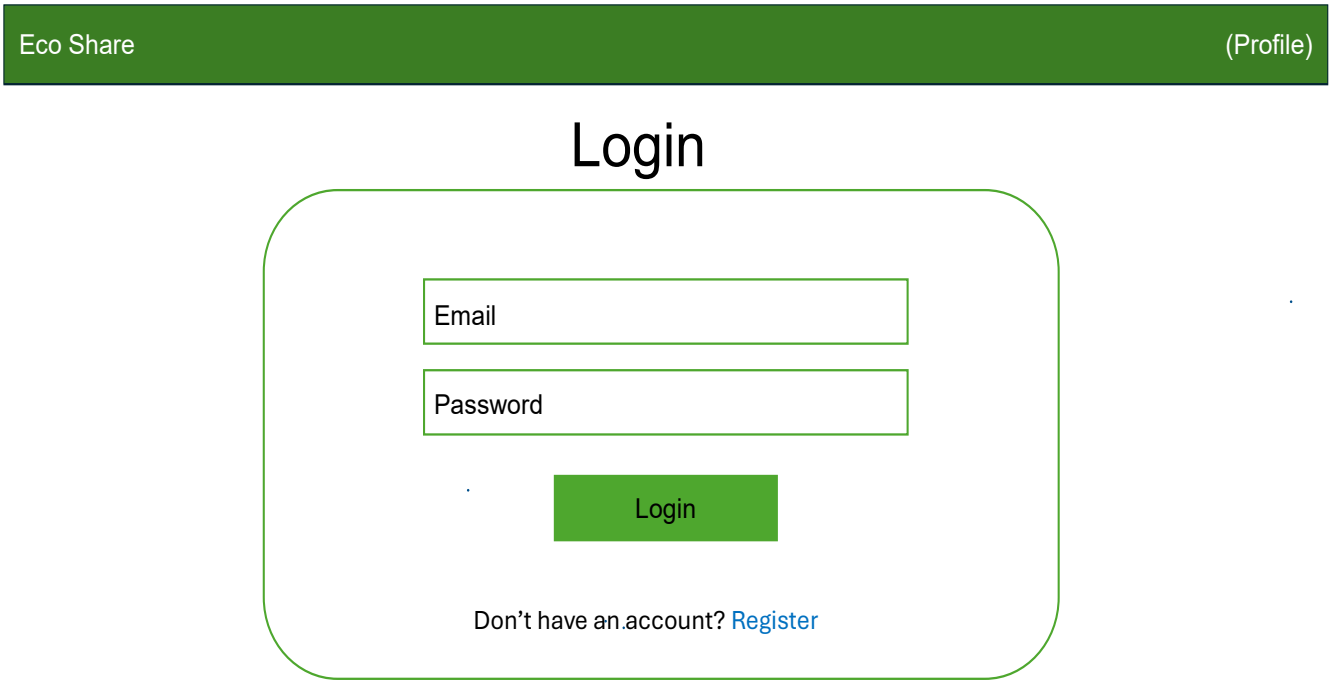
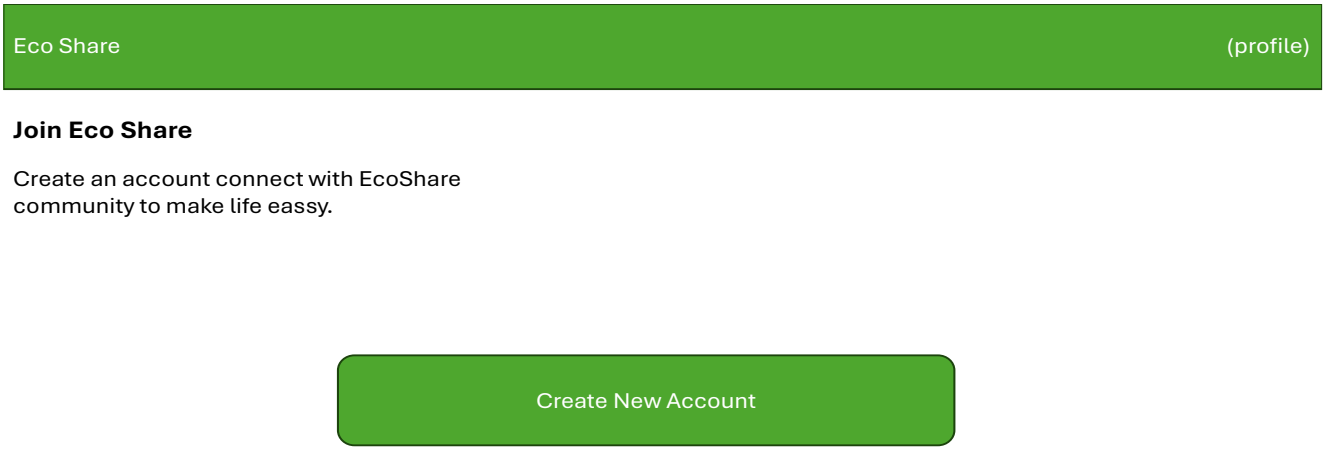


Figure 1: Login Page Wireframe
This login interface emphasizes simplicity by featuring minimal input fields, enhancing both usability and accessibility. Its centred layout helps lower cognitive load and provides clarity for users from all technical backgrounds.



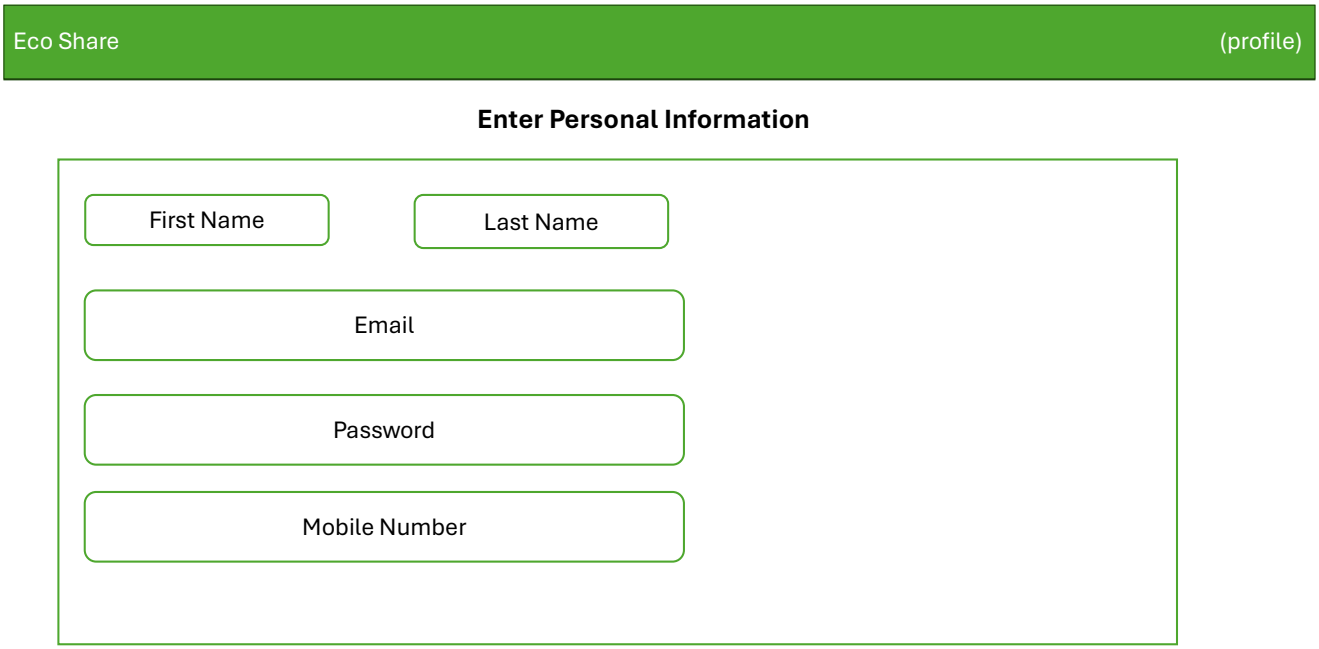


Figure 3 Create Account Screen Wireframe: This screen enables users to create an account by entering their personal information.

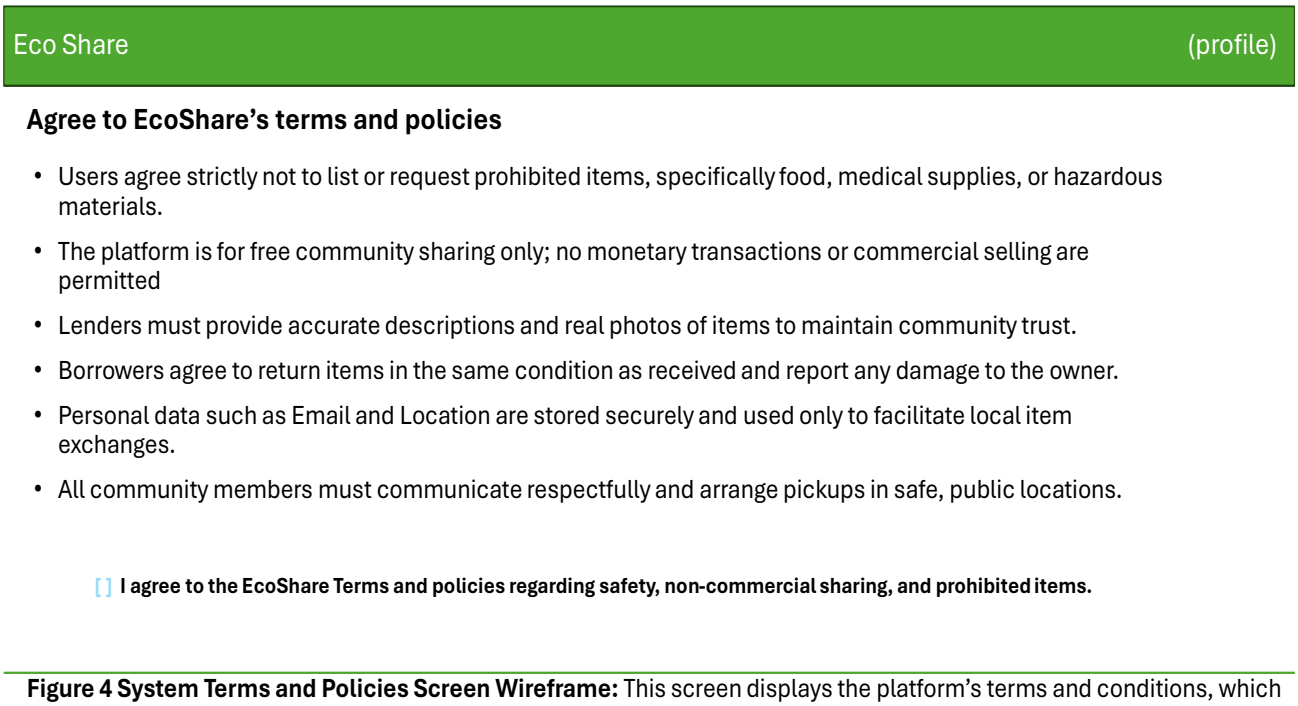


Figure 4 System Terms and Policies Screen Wireframe: This screen displays the platform's terms and conditions, which users must agree to before creating an account.

Eco Share (profile)

Enter the confirmation code

To confirm your account, enter the four-digit code we sent on SMS or email.

Confirmation Code

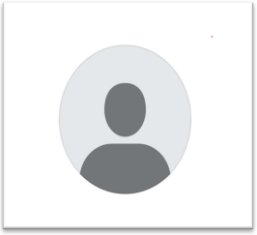
Next

I didn‘t receive the code

Figure 5 Confirmation Code Screen Wireframe: This screen enables users to enter a verification code to confirm their account and complete the registration process.

Eco share (profile)

Eco Share



Welcome to EcoShare --------

Figure 6 User Profile Screen Wireframe: This screen allows users to manage their account and access system features after logging in.

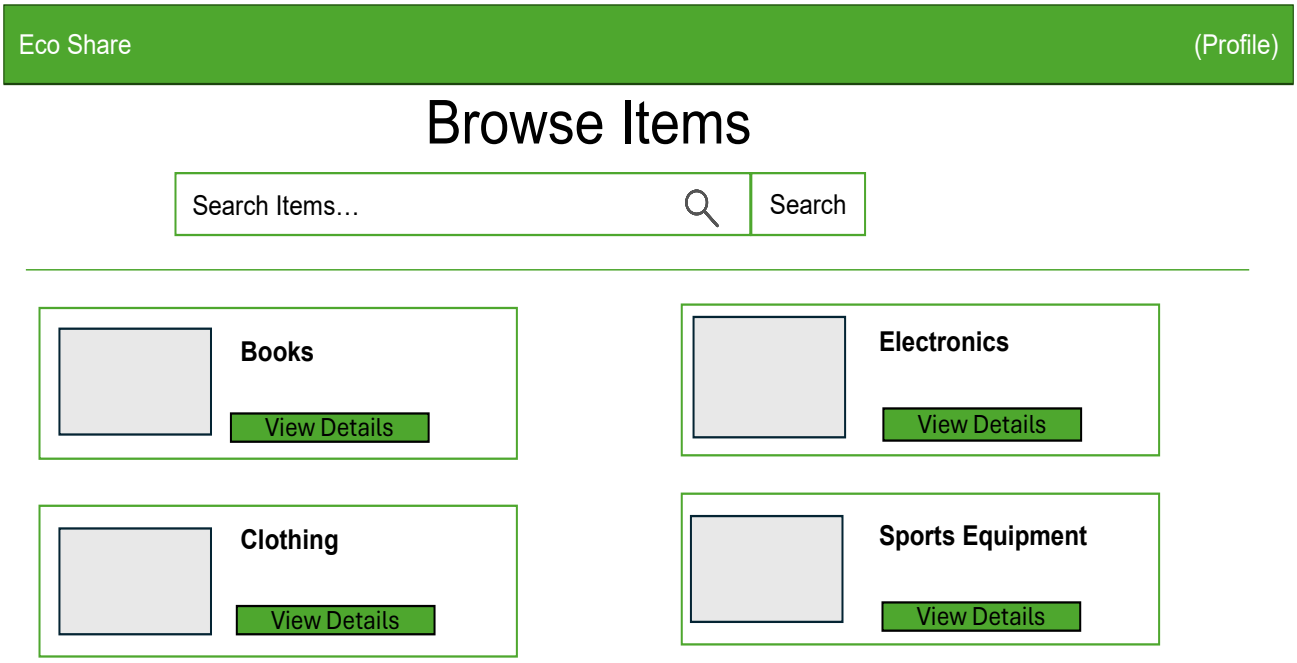


Figure 7: Browse Items Wireframe
The grid layout ensures clear organisation and easy navigation. Search functionality improves usability, while consistent item cards enhance accessibility and reduce cognitive load.

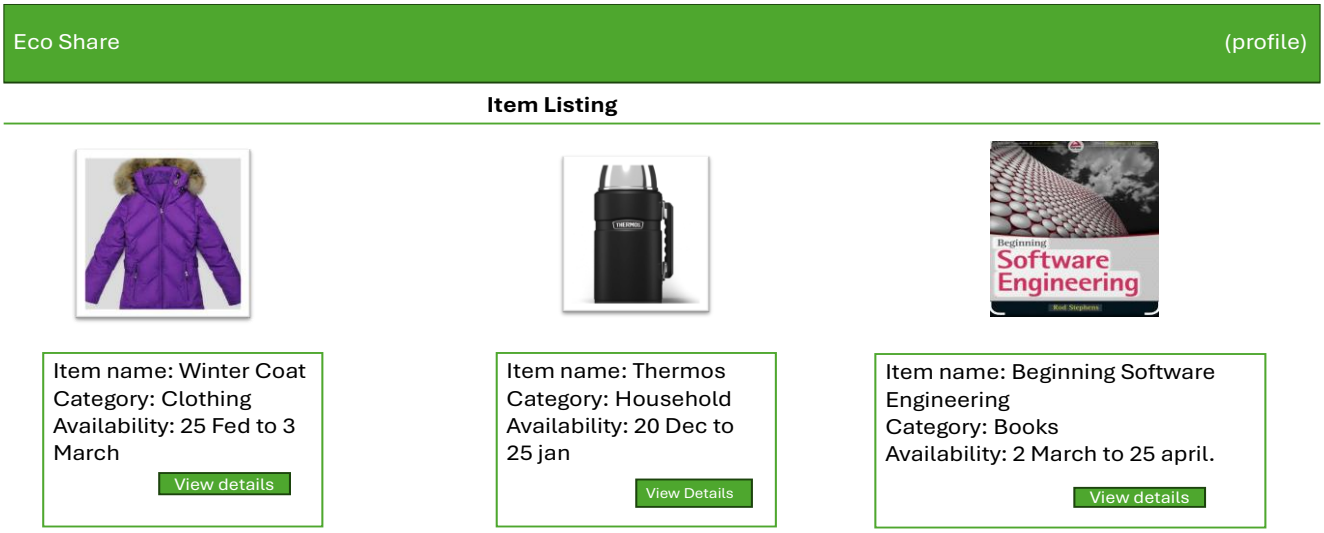


Figure 8: Browse Items Screen Wireframe
This screen displays available items in a structured grid layout. Users can search and filter items by category before selecting a specific item to view detailed information.

Eco Share (Salina)

Item Name: Iron

Iron

Owner: Salina

Category: Electronics

Availibility: 24 Feb – 5 March

Description

Request item

Figure 9 Item details Page Wireframe: The two-column layout separates image and information for improved readability. Clear structure and a visible request button enhance usability and support user interaction.

Eco Share (Profile)

Create New Listing

Item Title

Category

Location

Description

Upload Image

Submit Listing

Figure 10 Create new listing Wireframe: This create listing page uses a clear, structured form layout to ensure ease of use.

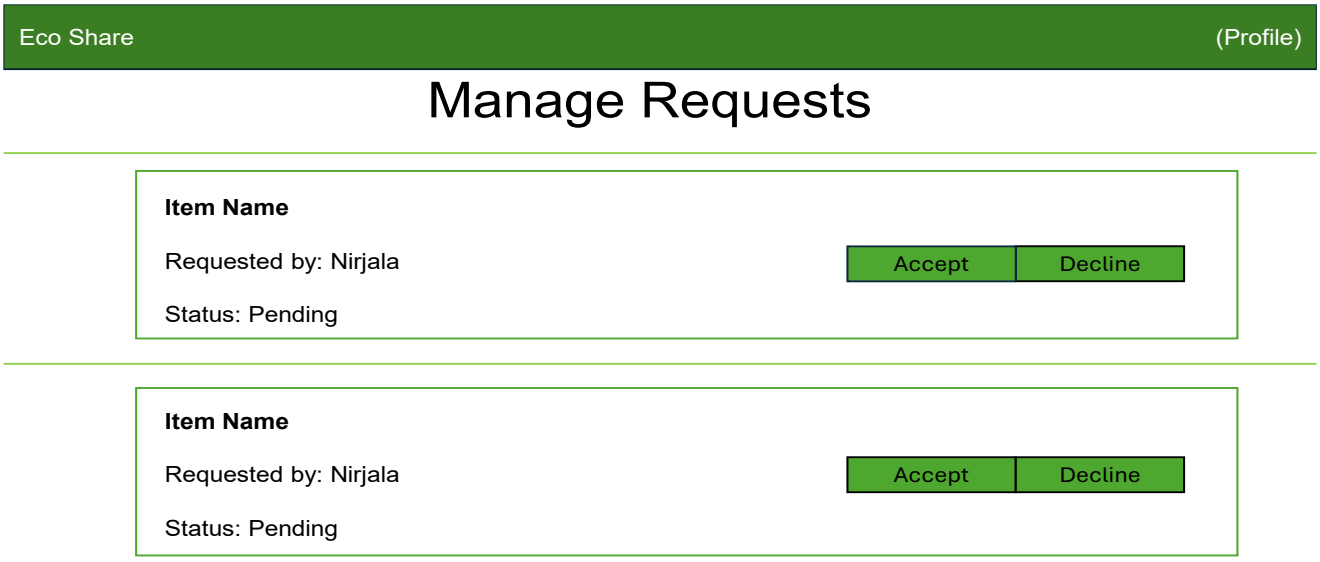


Figure 11 Request page Wireframe: The requests page presents information in a structured list format for clarity and easy management. This page represents incoming requests received by the item owner. The owner can accept or decline borrow requests, ensuring proper access control within the system.

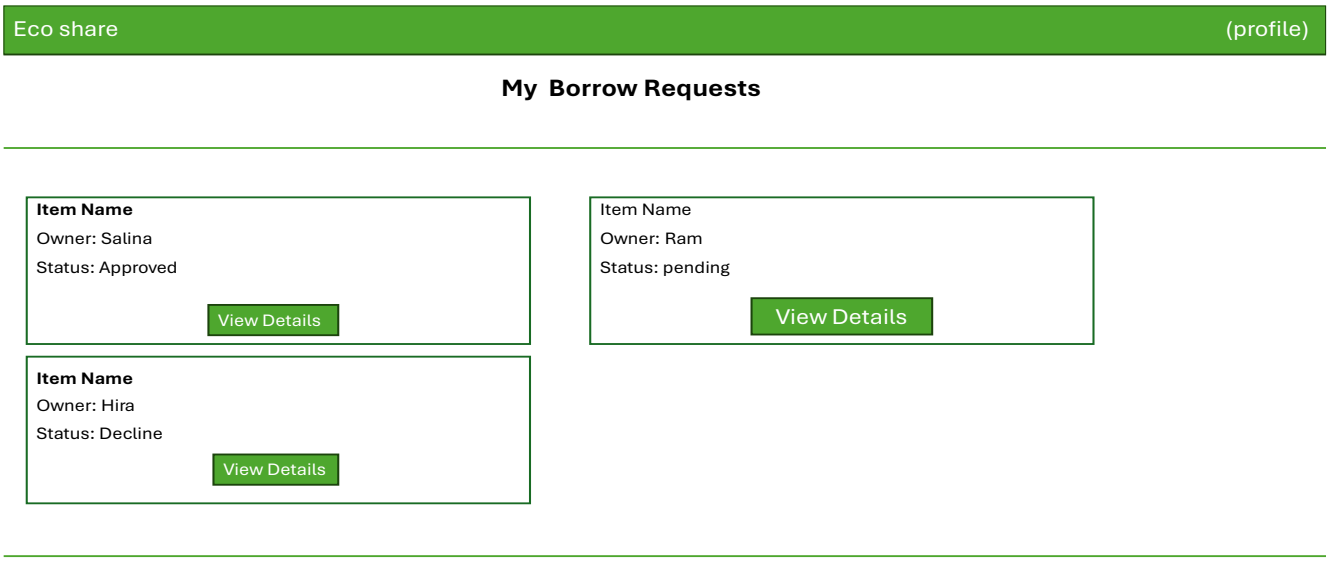


Figure 12 My Borrow Requests page Wireframe: This screen allows borrowers to view the status of their submitted requests, ensuring transparency and clear communication.

Eco Share (profile)

Approved Request Details

Iron

Owner: Salina

Pickup location: Greenford

Date: 26 Feburary

Message owner | Confirm Pickup

Figure 13 Approved Request Details page Wireframe: This page displays confirmed pickup details and communication options after a request has been approved. It enhances transparency and promotes user safety during item exchange.

UI Design Principles & Style Guide

Colour Strategy
•Light green used as the primary background colour to reflect sustainability

•Dark green used for headers and buttons to create contrast and clear visual hierarchy

•Colour consistency maintained across all screens

Layout Consistency
•Uniform header displayed on every page

•Consistent button design and placement

•Structured spacing to reduce cognitive load
•Grid layout used for browsing items

Typography
•Sans-serif font (e.g., Arial)

•Clean and modern design

•Enhances readability across different devices

Accessibility Considerations
•High contrast between light green and dark green elements

•Clear button labels and form fields

•Simple navigation structure

•Logical content hierarchy

Sustainability Alignment
The green colour palette reinforces the platform's sustainability objectives and strengthens visual identity. The simple and accessible interface encourages community participation in item reuse, supporting waste reduction and sustainable behaviour.

6. Activity Diagrams/ Flowcharts

This section presents flowcharts that describe the detailed process logic of key features within the Eco Share system. These flowcharts illustrate decision-making, user actions, and system behaviour to provide a deeper understanding of how the application operates.

i) Flowchart 1: User Registration and Login Process

This flowchart outlines the steps involved in user authentication, including registration, login, and validation.

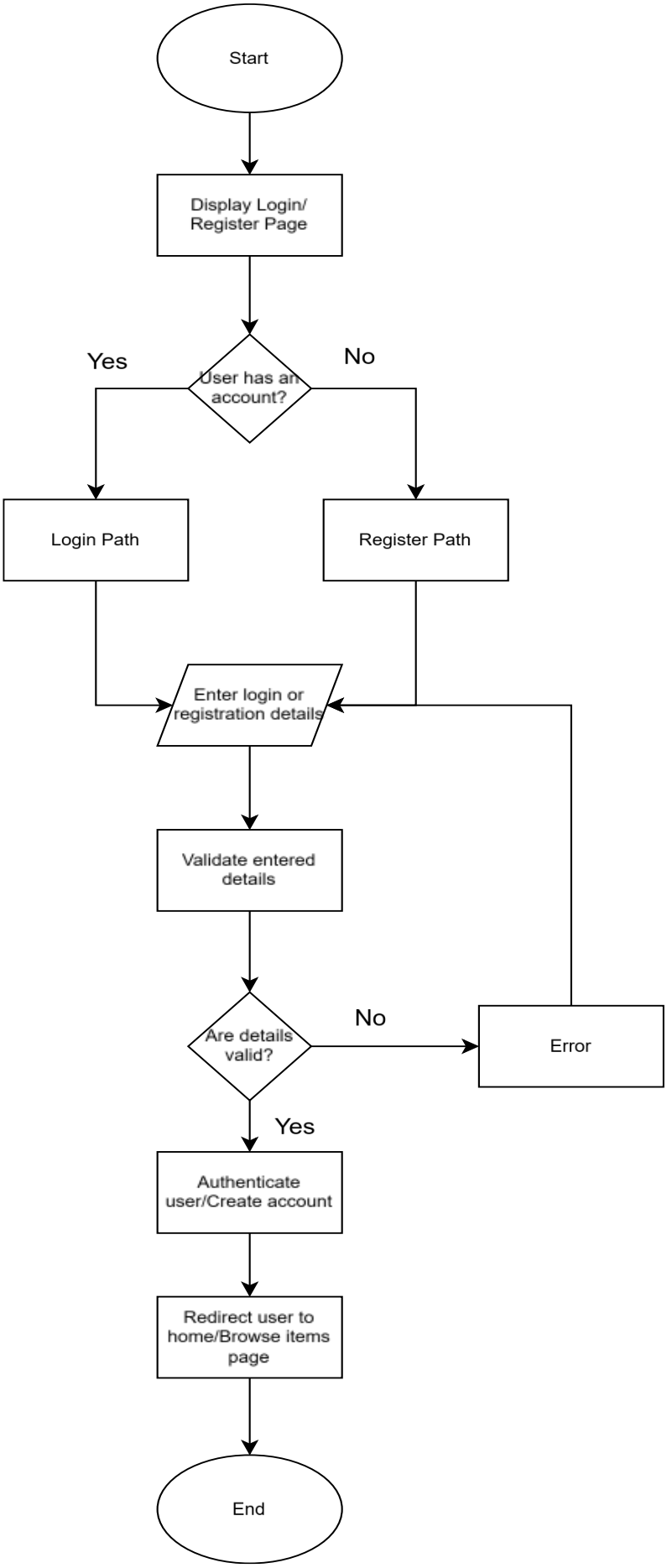


Figure 2: User Registration and Login Flowchart
The flowchart shows how users gain access to the system through secure authentication.

ii) Flowchart 2: Create Item Listing

This flowchart illustrates the process of creating a new item listing within the Eco Share platform.

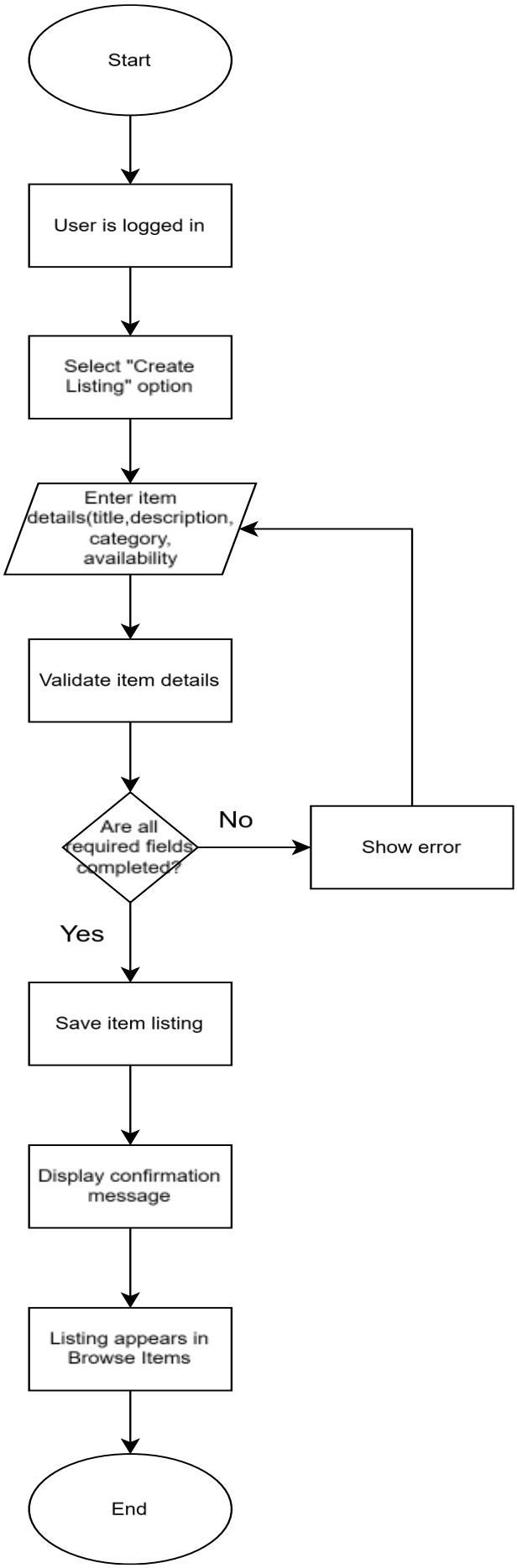


Figure 3: Create Item Listing Flowchart

The flowchart demonstrates how item details are validated and published for borrowing.

iii) **Flowchart 3:** Borrow Request, Approval, and Pickup

This flowchart describes the borrowing workflow within Eco Share, focusing on approval decisions, communication between users, and safety considerations.

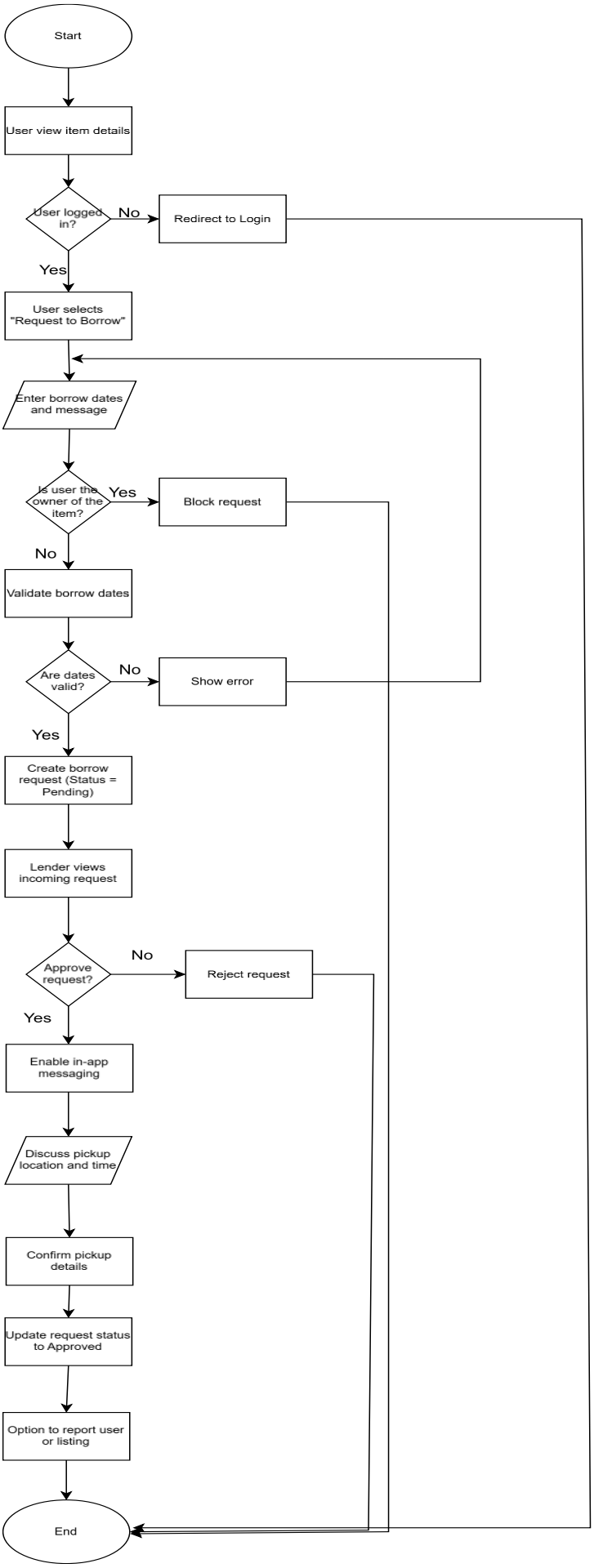

Start
User view item details
User logged in?
No
Redirect to Login
Yes
User selects "Request to Borrow"
Enter borrow dates and message
Is user the owner of the item?
Yes
Block request
No
Validate borrow dates
Are dates valid?
No
Show error
Yes
Create borrow request (Status = Pending)
Lender views incoming request
Approve request?
No
Reject request
Yes
Enable in-app messaging
Discuss pickup location and time
Confirm pickup details
Update request status to Approved
Option to report user or listing
End

Figure 4: *Borrow Request, Approval, and Pickup Flowchart*
This flowchart demonstrates how borrow requests are processed, approved, and completed using in-app messaging to agree on pickup details safely.

Activity diagram

The following activity diagram illustrates the borrowing workflow in Eco Share, highlighting interactions between the borrower, lender, and system.

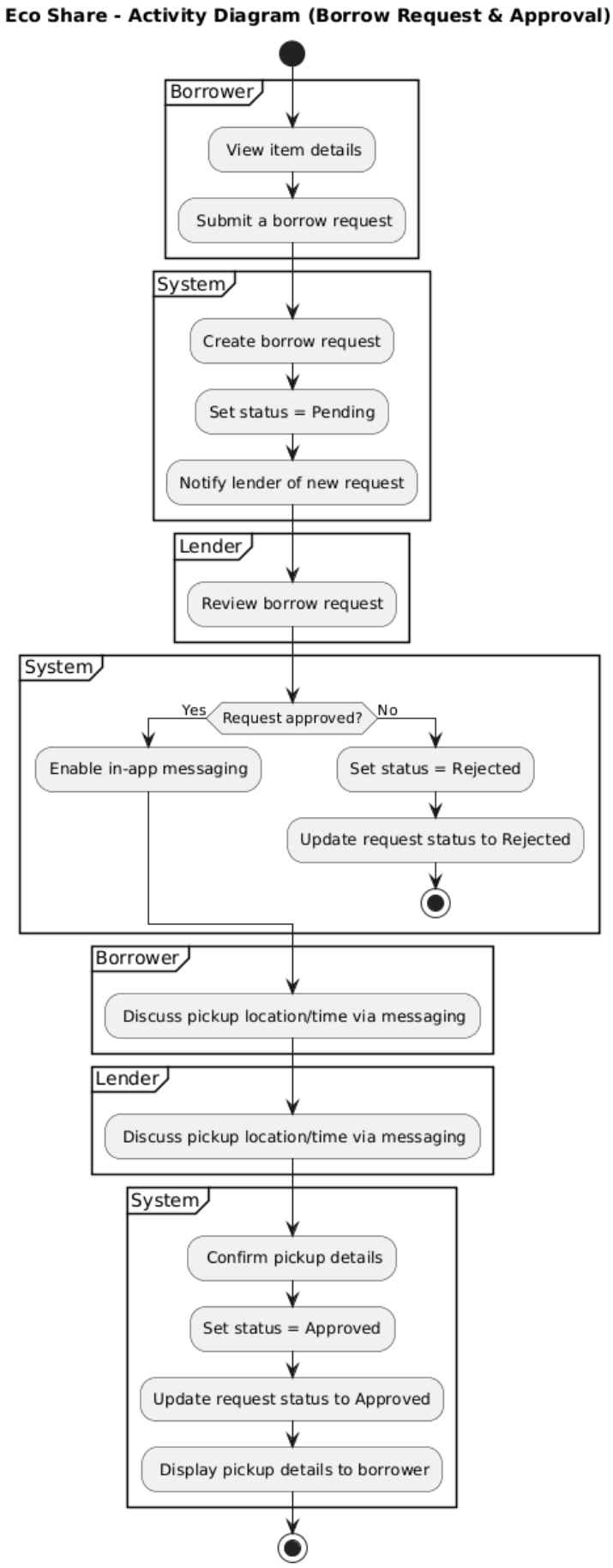


Figure 5: Activity Diagram – Borrow Request and Approval Process
This diagram shows how borrow requests are reviewed, approved, or rejected, and safely completed through in-app messaging.

7. Other Artefacts

This section presents additional design artefacts used to further support the understanding of the Eco Share system. An Entity Relationship Diagram (ERD) is included to illustrate the core data structure and relationships between users, items, borrow requests, messaging, and reports.

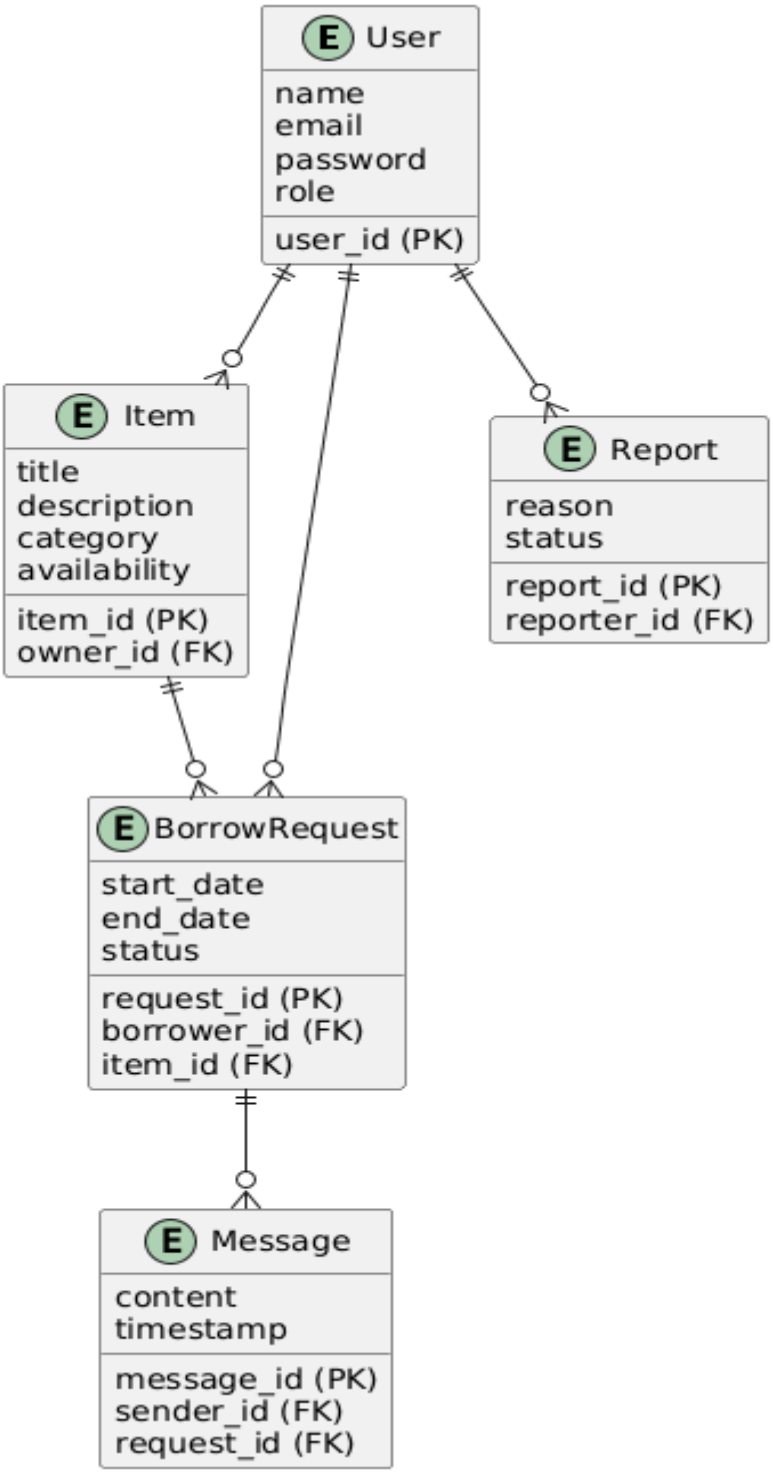


Figure 6: Entity Relationship Diagram (ERD) for the Eco Share system.

8. Sprint 3 Planning (Kanban Board)

Sprint 3 tickets were created based on the prioritised user stories and added to the Kanban board. Tasks were assigned to group members to ensure clear responsibility and readiness for implementation.

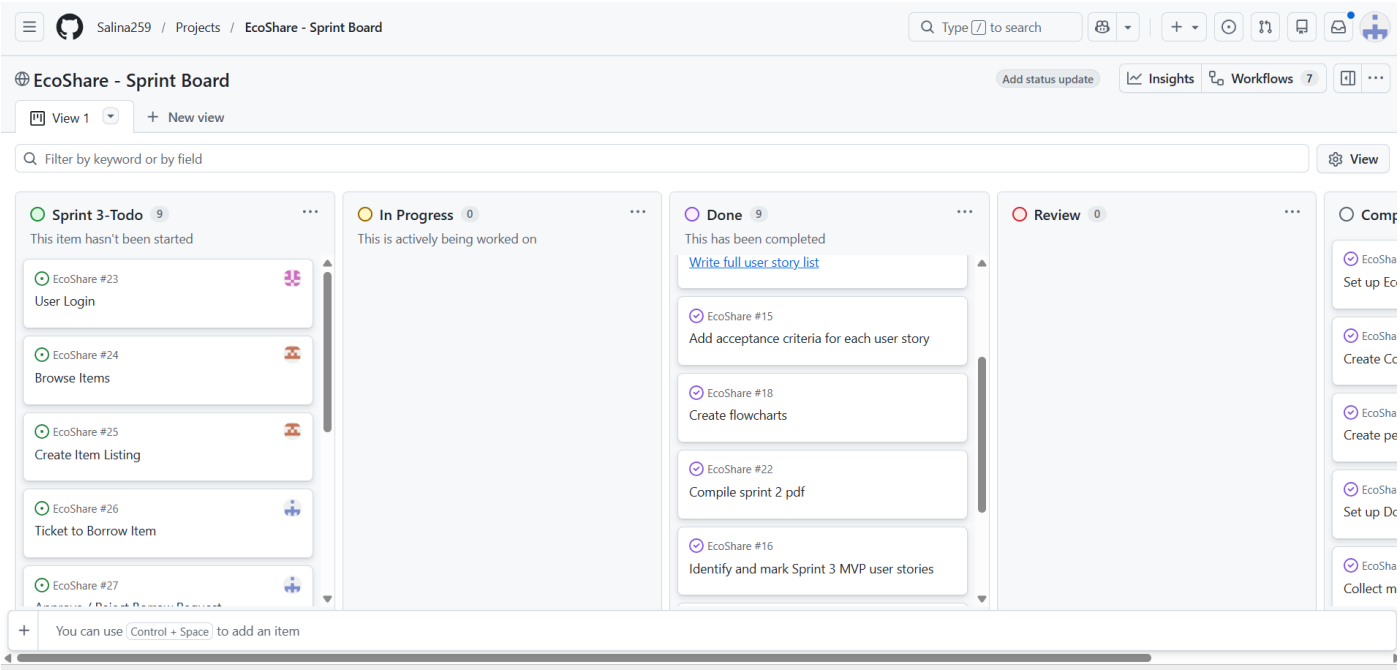


Figure 7: Sprint 3 Kanban Board with Assigned Tasks

9. Meeting records

- **First meeting record**

Meeting Minutes

Date and Time	18-02-2026 Wednesday, 1:00 pm -4:00 pm
Project Name	Eco Sharing Application
Meeting Goal	Assign Sprint 2 roles and Responsibilities
Facilitator	Salina
Note taker	Nirjala
Attendees	Salina, Nirjala, Hira, Aayush
Roundtable Updates (each group member to contribute)	**Hira** Started drafting user stories and acceptance criteria. **Salina** Updated Kanban board, reviewed GitHub structure, and pdf. **Nirjala** Wireframes in progress. **Aayush** planned use case diagram and flowcharts.
Discussion points	Kanban board update for sprint2. Personas were reviewed and aligned with user stories. Discussed privacy, fair use and safe item exchange. Each member will complete their assigned before the deadline.
Actions	All members work in progress.

In this meeting, the team planned Sprint 2 activities and assigned roles to each member. We discussed user stories, diagrams, wireframes, and updated the Kanban board. Key project requirements and safety policies were also reviewed to ensure everyone understood their responsibilities.

- **Second meeting record**

Meeting Minutes

Date & time	Monday, 23 February 2026, 11 am to 2 pm
Project Name	Eco Share Application
Meeting Goal	Final quality check of all requirements and merging sections into the final PDF.
Facilitator	Salina
Note taker	Nirjala
Attendees	Salina, Nirjala, Hira, Aayush
Roundtable Updates	**Salina** update the kanban bord, manage the GitHub README, and prepared the final submission file. **Nirjala** finalized all wireframes, including the register page and approved request screen with pickup details. **Hira** finalized the User Story list and the logic for the Report User. **Aayush** completed the Use Case and Class diagrams, ensuring they align with the frontend navigation flow.
Discussion points	The team verified that all mandatory UI features—specifically the Pickup Location, Time Window, and Report User safety button—were successfully integrated. We also confirmed the consistent Green/Arial visual style and finalized the mandatory policy prohibiting the listing of hazardous items or food to ensure community safety.
Actions	Nirjala, Aayush, and Hira complete their work and hand over to Salina. After that She merge all parts into one pdf and submit is for the team is in process.

This meeting focused on reviewing all Sprint 2 deliverables. Each member presented their completed work, and we confirmed that all MVP features were included. The team then prepared to merge everything into one final PDF for submission.

10. Kanban board link

https://github.com/users/Salina259/projects/2/views/1

11. GitHub Link

https://github.com/Salina259/EcoShare.git

13. Conclusion:

We focused on refining the system requirements and producing detailed design artefacts for the Eco Share application. During this sprint, the team finalised the project summary, prioritised user stories, and developed key design components, including use case diagrams, wireframes, activity diagrams, and the ERD. These artefacts helped clarify the system structure, user interactions, and overall workflow.

The Sprint 3 Kanban board was also prepared with assigned development tasks based on the MVP features. As a result, the project is now well-structured and ready to move into the implementation phase in Sprint 3.